



L.N. MALVIYA INFRA PROJECTS PVT. LTD.

Registered Office : Plot No. 29, F/F Sector-11, Dwarka, New Delhi - 110078

Corporate Office : T-10, IIIrd Floor, City Centre, Press Complex,

Plot No. 1, M.P. Nagar, Zone-I, Bhopal (M.P.) 462011,

Tel./Fax : 0755 - 4295421

E-MAIL : LNMALVIYA@LNINFRA.COM

CIN No. : U45201DL2010PTC373715

Mob.: 9826452711, 7773000871, 9977004686, 9617777944 | E-mail : lnmalviya@lninfra.com | Web.: www.lninfra.com

Date: - 4/10/2023

No. LNMIPPL/AR/NHAI/Ujjain-Garoth/2023-24/5697

To,

Authorized Representative

Ujjain Garoth Road Project Private Limited

Registered Office: - 1st Floor, AML, Centre,

8 Mahakali Caves Road, Andheri (East),

Mumbai -400093

Email: info.ghvgroup.com

Sub: 4- Lane of Ujjain Garoth (Package-I) Chandesari to Khedakhajuriya from Km. (-) 750 to 0+000 & 0+00 to 41.400 of NH-752D in the state of Madhya Pradesh for HAM Mode. Reg. Review of Shop Drawing for Elastomeric Bearing from Source Polymer Products along with Quality Assurance Plan (QAP) for Approval.

Ref: Concessionaire letter no. GHV/NHAI/UG-PKG1/2023-24/514 Dated-18.09.2023

Dear Sir,

The Drawing for Elastomeric Bearing received through email have been reviewed and seems to be in order.

Loads may please be cross verified from approved design and drawings to avoid any mistake.

If there is any mismatch as per Concession Agreement, responsibility lies with "M/S Ujjain Garoth Road Project Private Limited" to rectify the same at their own cost.

Thanks and regards

For, L. N. Malviya Infra Projects Pvt. Ltd.


(Authorized Signatory)

Copy to:

1. The Project Director, NHAI, PIU, Ujjain
2. The Team Leader, L. N. Malviya Infra Projects Pvt. Ltd., Ujjain
Email: lnm.ujjaingarothpkg1@gmail.com
3. L. N. Malviya Infra Projects Pvt. Ltd, Bhopal

UJJAIN GAROTH ROAD PROJECT PRIVATE LIMITED

GHV/NHAI/UG-PKG1/2023-24/514

Date-18.09.2023

To
Team Leader
L N Malviya Infra Projects Pvt Ltd
In association with Intratech Civil Solution & Consultant
Plot No. -87/04, Subhas Marg, Tehsil Barod,
Agar Malwa (M.P.)-465550

File

Sub :- 4-Lane of Ujjain-Garoth (Package-I) Chandesari to Khedakhajuriya from Km. (-) 750 to 0+000 & 0+00 to 41.400 of NH-752D in the State of Madhya Pradesh for HAM Mode - **Regd. Submission of Shop Drawing for Elastomeric Bearing from Source Polymer Products along with Quality Assurance Plan (QAP) for Approval regd**

Ref.:-1. Contract Agreement Date 05.05.2022

2. Appointment Date 29.09.2022

3. IE's Letter No.LNMIPPL/AR/NHAI/Ujjain-Garoth/2023-24/2394 on dated 19.06.2023

4. IE's Letter No.LNMIPPL/AR/NHAI/Ujjain-Garoth/2023-24/2736 on dated 01.07.2023

Dear Sir,

In pursuant to clause no.12.2 of the Contract Agreements, we are submitting herewith Shop Drawing for Elastomeric Bearing from Source **Polymer Products** along with Quality Assurance Plan (QAP) for above said project.

This is submitted for your kind review and necessary approval please.

Thanking you and assuring you of our best services at all times.

With Regards

[Signature]
Authorized Signatory

For , Ujjain-Garoth Road Project Private Limited

[Signature]
[Signature]

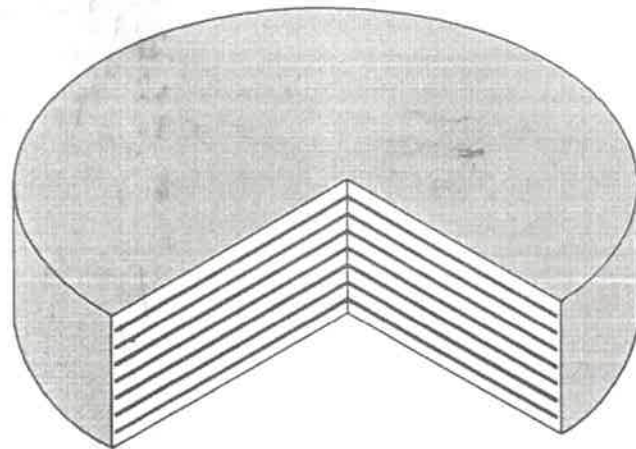
Enclosures: As above

Copy to:- 1. Project Director, NHAI, Project Implementation Unit(PIU), Ujjain
2. GHV (India) Pvt Ltd , Base Camp, Panvihar, Ujjain.

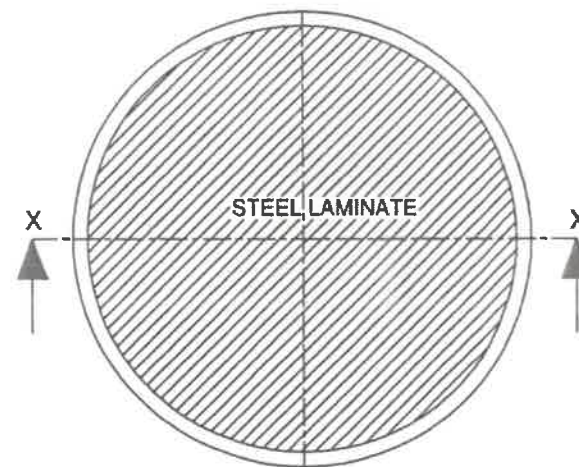
POLYMER PRODUCTS

SHOP DRAWING FOR ELASTOMERIC BEARING

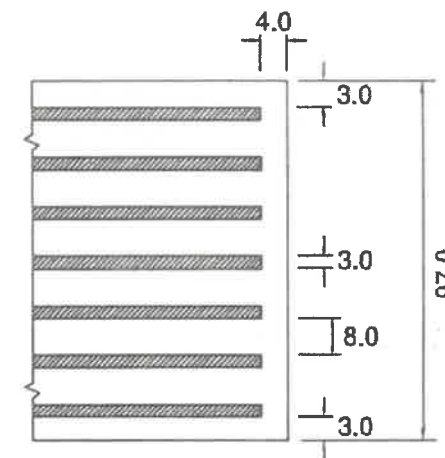
(CONFORMING TO IRC : 83 (PART-II) 2018 SPECIFICATION)



ISOMETRIC VIEW OF CIRCULAR ELASTOMERIC BEARING
THE NUMBER OF STEEL LAMINATES AND ELASTOMERS ARE INDICATIVE ONLY. THE NUMBER SHALL NOT BE COUNTED, ONLY THE PROVIDED INFORMATION IN THE SECTION X - X SHALL BE REFERRED



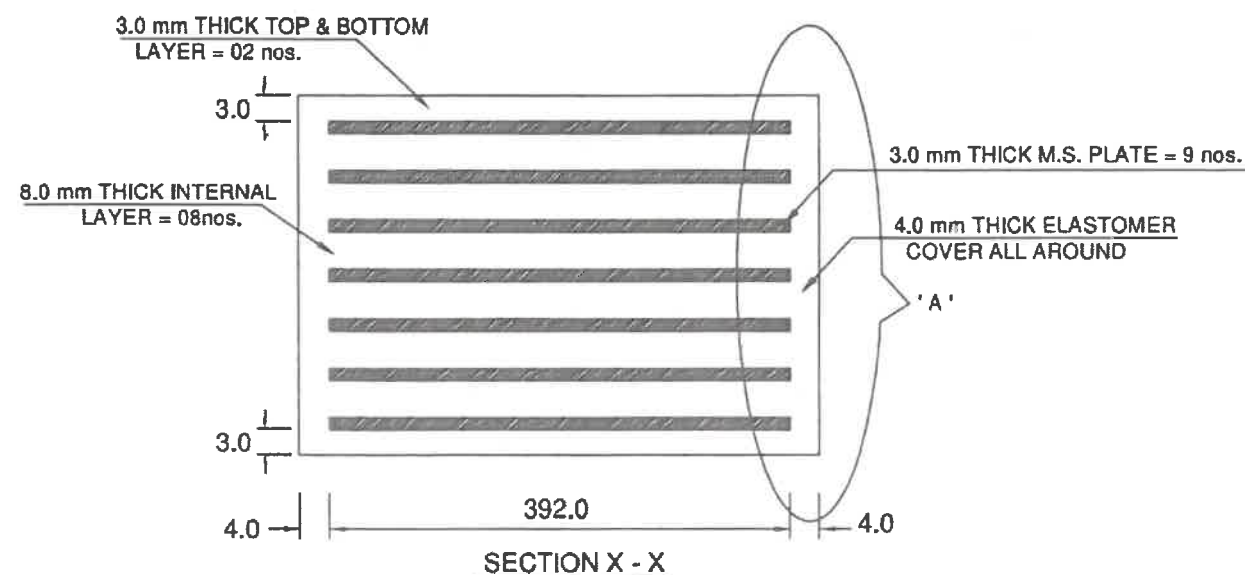
PLAN OF ELASTOMERIC BEARING
(PLAN SHOWING STEEL LAMINATE)



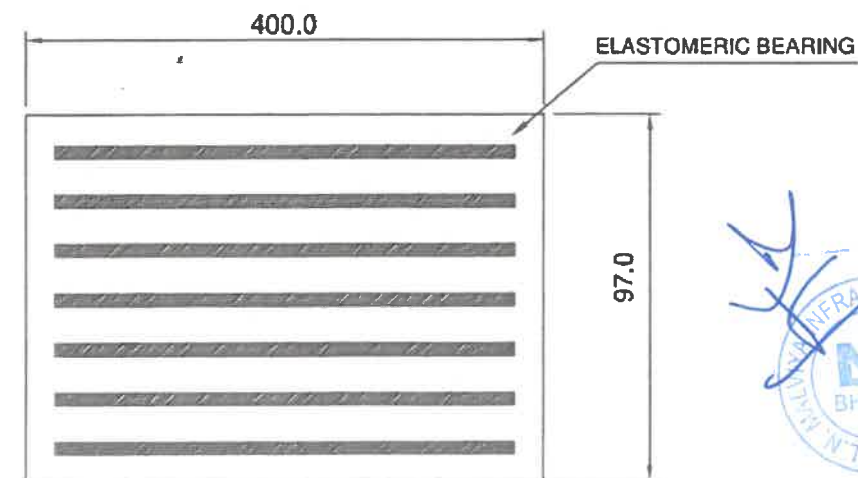
DETAILS 'A'

NOTES :-

- ALL DIMENSIONS ARE IN MILLIMETERS UNLESS OTHERWISE MENTIONED.
- NO DIMENSIONS SHALL BE SCALED FROM THIS DRG. ONLY WRITTEN DIMENSIONS SHALL BE FOLLOWED.
- THE BEARING SHALL BE FULLY MOULDED WITH STEEL LAMINATIONS IN BETWEEN AS SHOWN.
- THE EDGES OF PLATE SHALL BE SLIGHTLY ROUNDED OFF OR CHAMFERED.
- SPECIFICATION INCLUDING PERMISSIBLE TOLERANCES FOR THE ELASTOMERIC BEARINGS SHALL BE IN ACCORDANCE WITH IRC : 83 (PART II) - 2018 STANDARD SPECIFICATION AND CODE PRACTICE FOR ROAD BRIDGES SECTION IX PART II ELASTOMERIC BEARINGS.
- MATERIALS:
6.1 NATURAL RUBBER (NR) SHALL BE RSS 1X CONFORMING TO IS: 15361 : CONFORMING TO IRC : 83 (PART II) - 2018 WITH SHEAR MODULUS OF THE ELASTOMER BEARING SHALL BE 1.15 MPa.
- SPECIFICATIONS FOR MANUFACTURE , TESTING , ACCEPTANCE INSTALLATION AND MAINTENANCE OF BEARING SHALL BE AS PER IRC : 83 (PART II) - 2018.
- BEARINGS SHALL BE HANDLED CAREFULLY. THESE SHALL BE PROTECTED FROM BRIGHT SUN LIGHT AND EXTREME COLD. THESE SHALL BE STORED NEATLY UNDER COVER TILL INSTALLATION.



SECTION X - X



ARRANGEMENT OF BEARING

SHOP DRAWING AND QUANTITY ARE AS PER BELOW DETAILS :
ORDERED QUANTITY = 220 Nos.

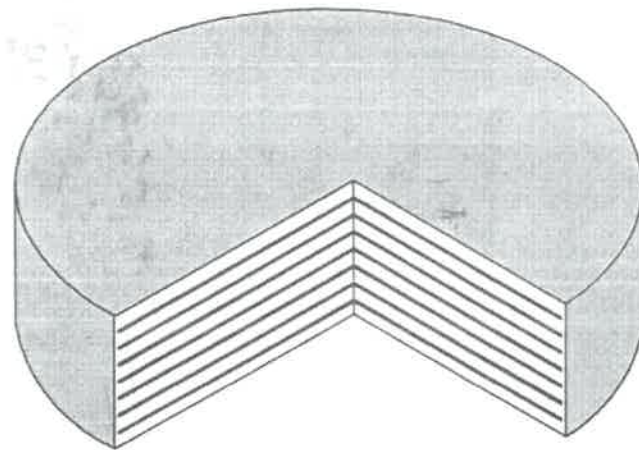


PROJECT	4-LANING OF UJAIN- GARDTH (PACKAGE-1) CHANDESARI TO KHEDAKHA JURIY A FROM Km 0+000 TO Km (-) 0+750 & Km 0+000 TO KM 414400 SECTION OF NH-148NG IN STATE OF MADHYA PRADESH ON HYBRID ANNUITY MODE				
CONTRACTOR	GHV (INDIA) PRIVATE LIMITED				
TITLE	DETAILS OF (400 DIA x 97) mm - ELASTOMERIC BEARING				
 POLYMER PRODUCTS Corporate Off. : 72, Mahaveer Centre, Sector - 17 Vashi, Navi Mumbai - 400 705 Tel : (022) 2799 1313 / 2528 Fax : 2788 1186 <u>Works</u> : H-32, MIDC, SATPUR, NASEK - 422 007 Tel. : (0253) 680461 Fax : 2350935 email : polymerproduct@rediffmail.com					
SCALE	REVISION	DRAWN	CHECKED BY	DATE	DRG. NO.
N. T. S.	Ro	T. J. ATHAN	S KOTGA	18/03/2023	PP / GHV_CIRCULAR / 400 DIA X 97

POLYMER PRODUCTS

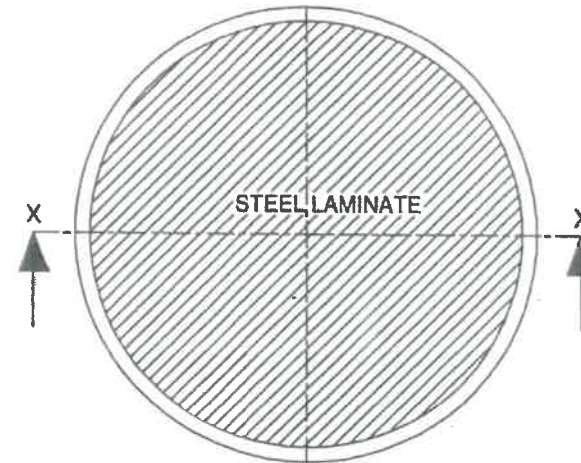
SHOP DRAWING FOR ELASTOMERIC BEARING

(CONFORMING TO IRC : 83 (PART-II) 2018 SPECIFICATION)

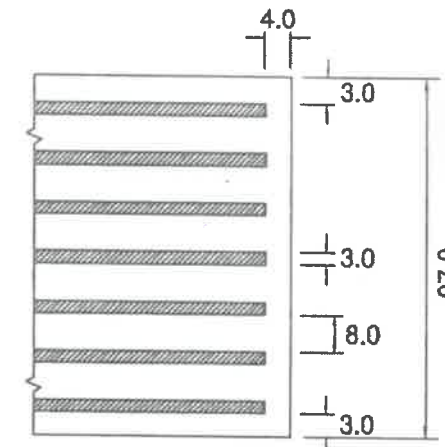


ISOMETRIC VIEW OF CIRCULAR ELASTOMERIC BEARING

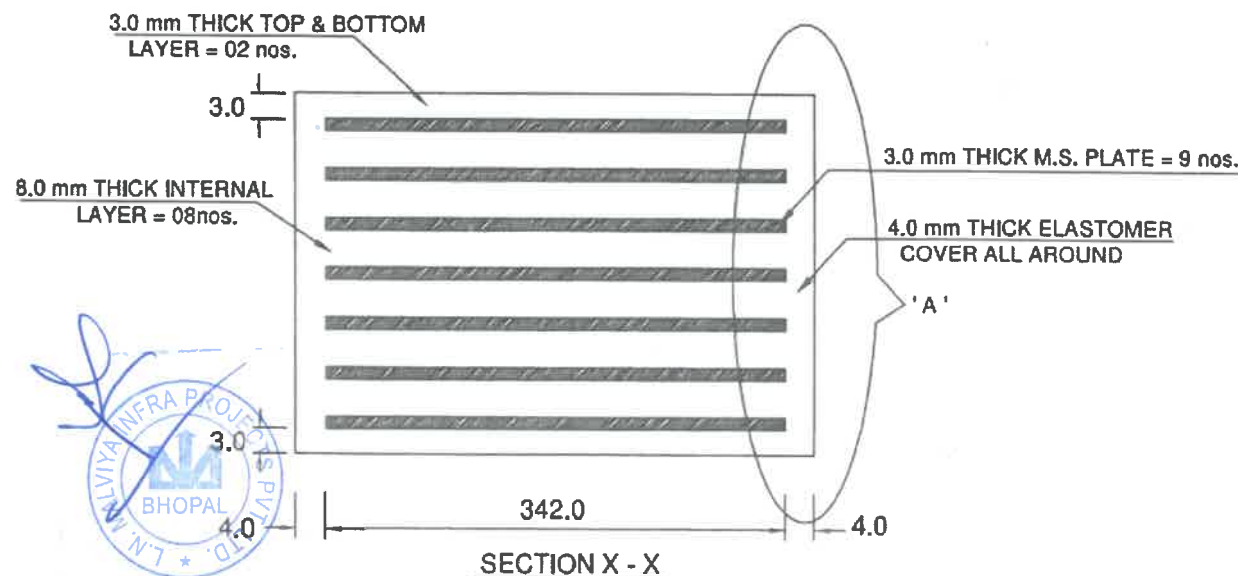
THE NUMBER OF STEEL LAMINATES AND ELASTOMERS ARE INDICATIVE ONLY. THE NUMBER SHALL NOT BE COUNTED, ONLY THE PROVIDED INFORMATION IN THE SECTION X - X SHALL BE REFERRED



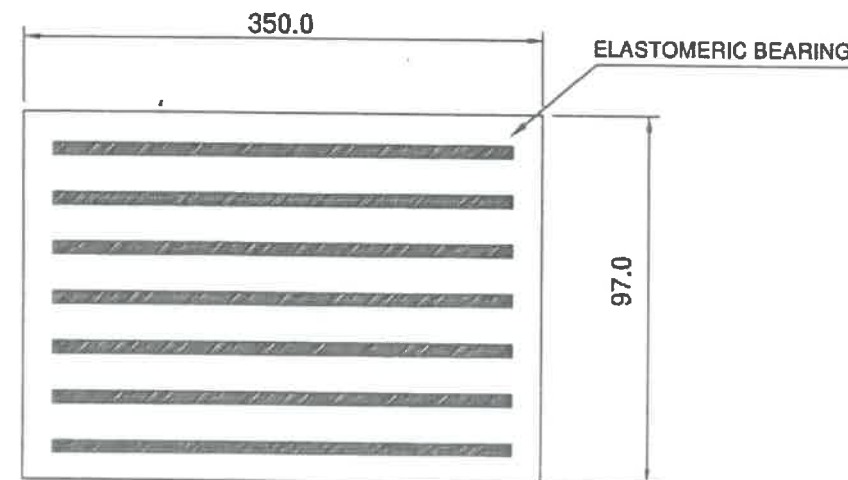
PLAN OF ELASTOMERIC BEARING
(PLAN SHOWING STEEL LAMINATE)



DETAILS 'A'



SECTION X - X



ARRANGEMENT OF BEARING

NOTES :-

- ALL DIMENSIONS ARE IN MILLIMETERS UNLESS OTHERWISE MENTIONED.
- NO DIMENSIONS SHALL BE SCALED FROM THIS DRG. ONLY WRITTEN DIMENSIONS SHALL BE FOLLOWED.
- THE BEARING SHALL BE FULLY MOULDED WITH STEEL LAMINATIONS IN BETWEEN AS SHOWN.
- THE EDGES OF PLATE SHALL BE SLIGHTLY ROUNDED OFF OR CHAMFERED.
- SPECIFICATION INCLUDING PERMISSIBLE TOLERANCES FOR THE ELASTOMERIC BEARINGS SHALL BE IN ACCORDANCE WITH IRC : 83 (PART II) - 2018 STANDARD SPECIFICATION AND CODE PRACTICE FOR ROAD BRIDGES SECTION IX PART II ELASTOMERIC BEARINGS.
- MATERIALS:
6.1 NATURAL RUBBER (NR) SHALL BE RSS 1X CONFORMING TO IS: 15361 : CONFORMING TO IRC : 83 (PART II) - 2018 WITH SHEAR MODULUS OF THE ELASTOMER BEARING SHALL BE 1.15 MPa.
- SPECIFICATIONS FOR MANUFACTURE , TESTING , ACCEPTANCE INSTALLATION AND MAINTENANCE OF BEARING SHALL BE AS PER IRC : 83 (PART II) - 2018.
- BEARINGS SHALL BE HANDLED CAREFULLY. THESE SHALL BE PROTECTED FROM BRIGHT SUN LIGHT AND EXTREME COLD. THESE SHALL BE STORED NEATLY UNDER COVER TILL INSTALLATION.

SHOP DRAWING AND QUANTITY ARE AS PER BELOW DETAILS :

ORDERED QUANTITY = 276 Nos.

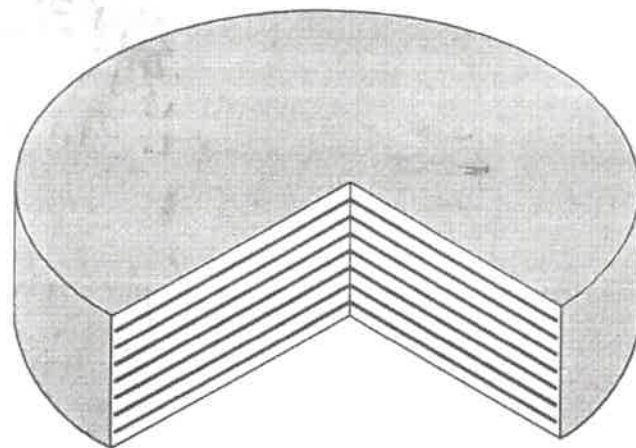


PROJECT	4-LANING OF UJJAIN- GAROTH (PACKAGE-1) CHANDESARI TO KHEDEKHA JURIY A FROM Km 0+000 TO Km (-) 0+750 & Km 0+000 TO KM 414400 SECTION OF NH-148NG IN STATE OF MADHYA PRADESH ON HYBRID ANNUITY MODE				
CONTRACTOR	GHV (INDIA) PRIVATE LIMITED				
TITLE	DETAILS OF (350 DIA x 97) mm - ELASTOMERIC BEARING				
	POLYMER PRODUCTS Corporate Office : 72, Mahavir Centre, Sector - 17 Vashi, Navi Mumbai - 400 705 Tel : (022) 2789 1310 / 2528 Fax : 2789 1198 Works : H-32, MIDC, SATPUR, NASHIK - 422 007 Tel. : (0253) 6604461 Fax - 2350935 email : polymerproduct@rediffmail.com				
SCALE	REVISION	DRAWN	CHECKED BY	DATE	DRG. NO.
N.T.S.	RD	T. J. ATHAN	S XOTGI	16/09/2023	PP / GHV_CIRCULAR / 350 DIA X 97

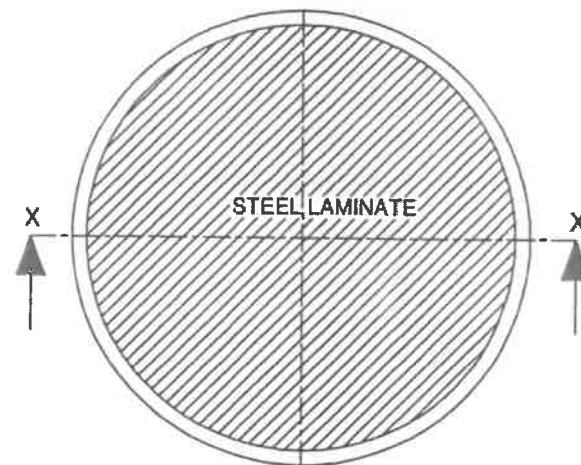
POLYMER PRODUCTS

SHOP DRAWING FOR ELASTOMERIC BEARING

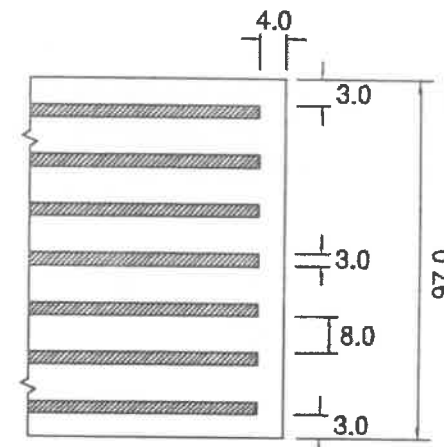
(CONFORMING TO IRC : 83 (PART-II) 2018 SPECIFICATION)



ISOMETRIC VIEW OF CIRCULAR ELASTOMERIC BEARING
THE NUMBER OF STEEL LAMINATES AND ELASTOMERS ARE INDICATIVE ONLY. THE NUMBER SHALL NOT BE COUNTED, ONLY THE PROVIDED INFORMATION IN THE SECTION X - X SHALL BE REFERRED



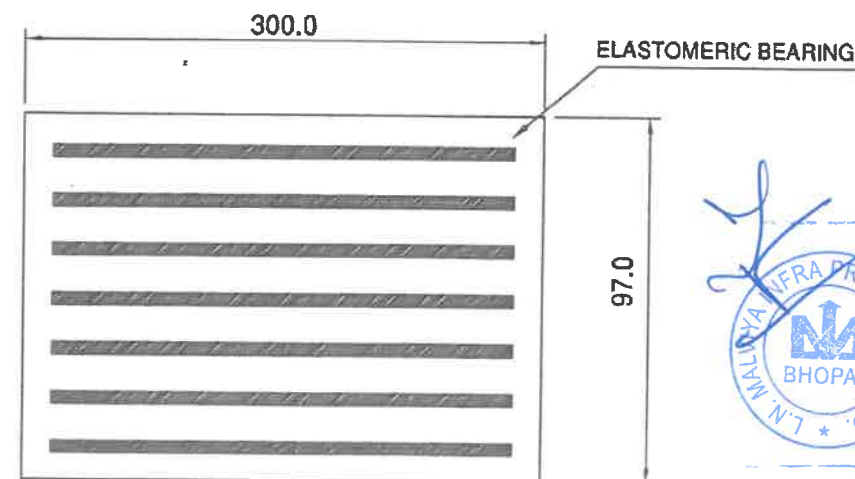
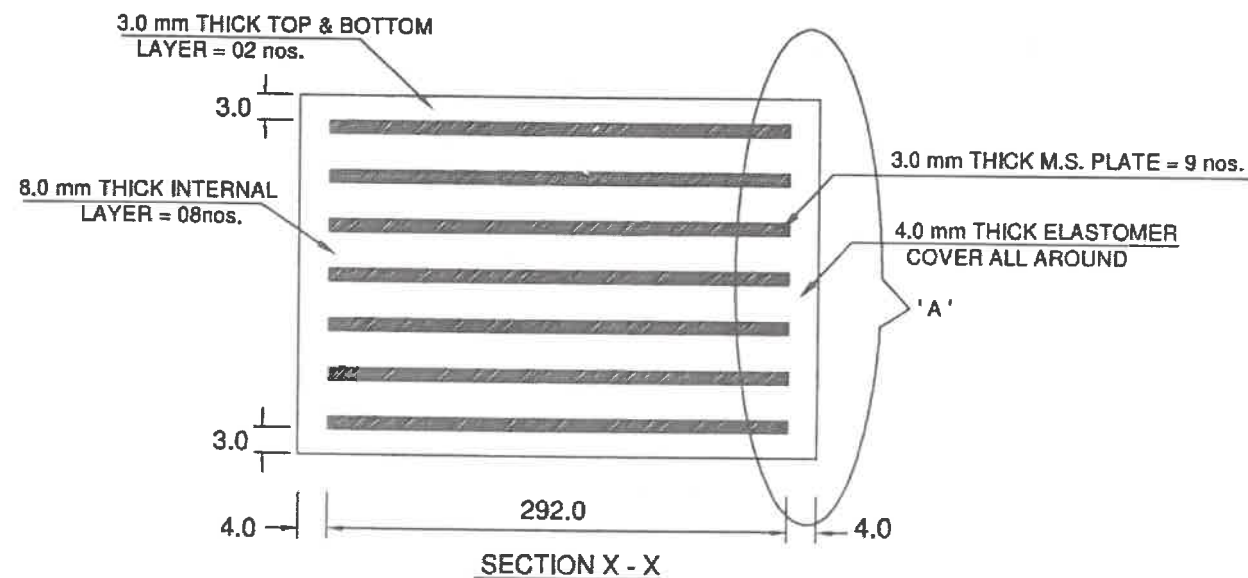
PLAN OF ELASTOMERIC BEARING
(PLAN SHOWING STEEL LAMINATE)



DETAILS 'A'

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETERS UNLESS OTHERWISE MENTIONED.
2. NO DIMENSIONS SHALL BE SCALED FROM THIS DRG. ONLY WRITTEN DIMENSIONS SHALL BE FOLLOWED.
3. THE BEARING SHALL BE FULLY MOULDED WITH STEEL LAMINATIONS IN BETWEEN AS SHOWN.
4. THE EDGES OF PLATE SHALL BE SLIGHTLY ROUNDED OFF OR CHAMFERED.
5. SPECIFICATION INCLUDING PERMISSIBLE TOLERANCES FOR THE ELASTOMERIC BEARINGS SHALL BE IN ACCORDANCE WITH IRC : 83 (PART II) - 2018 STANDARD SPECIFICATION AND CODE PRACTICE FOR ROAD BRIDGES SECTION IX PART II ELASTOMERIC BEARINGS.
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8. BEARINGS SHALL BE HANDLED CAREFULLY. THESE SHALL BE PROTECTED FROM BRIGHT SUN LIGHT AND EXTREME COLD. THESE SHALL BE STORED NEATLY UNDER COVER TILL INSTALLATION.



ARRANGEMENT OF BEARING



SHOP DRAWING AND QUANTITY ARE AS PER BELOW DETAILS :

ORDERED QUANTITY = 80 Nos.

PROJECT	4-LANING OF UJJAIN- GAROTH (PACKAGE-1) CHANDESARI TO KHEDAKHA JURIY A FROM Km 0+000 TO Km (-) 0+750 & Km 0+000 TO Km 414400 SECTION OF NH-148NG IN STATE OF MADHYA PRADESH ON HYBRID ANNUITY MODE				
CONTRACTOR	GHV (INDIA) PRIVATE LIMITED				
TITLE	DETAILS OF (300 DIA x 97) mm - ELASTOMERIC BEARING				
 POLYMER PRODUCTS Sompoti, Dist. : 72, Mahavir Centre, Sector - 17 Vashi, Navi Mumbai - 400 705 Tel : (022) 2789 1310 / 2828 Fax : 2789 1186 Work : H-32, MIDC, SATPUR, NASHIK - 422 007 Tel : (0253) 8604461 Fax : 2350935 email : polymerproducts@rediffmail.com					
SCALE	REVISION	DRAWN	CHECKED BY	DATE	DRG. NO.
N. T. S.	RD	T. J. ATHAN	S KOTGI	18/08/2023	PP / GHV_CIRCULAR / 300 DIA X 97